

The Acute Effect of a Diaphragmatic Breathing Warm-Up on Maximal Deadlift Performance

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Abstract

Katz, NS, Rossow, LM, and Fahs, CA. The acute effect of a diaphragmatic breathing warm-up on maximal deadlift performance. *J Strength Cond Res* XX(X): 000–000, 2026—The purpose of this study was to investigate the effect of acute diaphragmatic breathing practice on muscular strength (deadlift 1 repetition maximum) performance. Forty resistance-trained adults (32 men, 8 women; age 24.8 ± 6.8 years) completed 2 testing sessions in a randomized crossover design. Each session involved a standardized warm-up followed by a 1-repetition maximum (1RM) deadlift assessment. In the experimental condition, subjects performed a 5-minute diaphragmatic breathing protocol in the supine position before testing. Deadlift 1RM was significantly greater in the diaphragmatic breathing condition than control (163.6 ± 47.1 vs. 160.7 ± 49.6 kg; mean difference = 2.9 kg, +2.8%; $p < 0.05$). The effect size was small (Cohen's $d = 0.34$). Twenty-four subjects (60%) demonstrated improved performance (mean +8.1 kg, +6.6%), while 16 (40%) subject showed no change or a decrease in 1RM (mean -4.8 kg, -2.8%). A brief diaphragmatic breathing warm-up produced a statistically significant but small improvement in maximal deadlift performance. Given its simplicity, minimal time requirement, and the fact it requires basic equipment (e.g., a bench or box for foot support and a foam roller placed between the knees), diaphragmatic breathing may serve as a practical warm-up strategy for athletes and lifters seeking marginal gains in maximal strength performance.

Key Words: strength, intra-abdominal pressure, trunk stability, resistance training, 90/90 breathing

Introduction

Maximal strength is essential for athletic performance and resistance training, and the deadlift is widely recognized as a multi-joint exercise requiring coordinated action of several large muscle groups, making it a common measure of whole-body strength (2). The ability to generate maximal force in this movement is influenced not only by muscular strength but also by trunk stability, which provides a foundation for efficient force transmission between the hip and shoulder complexes.

The diaphragm plays a central role in trunk stabilization through its contribution to intra-abdominal pressure (IAP). Prior research has demonstrated that diaphragmatic activity enhances postural stability and supports force transmission during isometric stabilization tasks that involve the trunk and proximal musculature (4). Diaphragmatic breathing has been described as the foundation of core stability (7), and breathing-focused interventions have been associated with improvements in postural stability and balance (11). Despite this evidence, there is limited research directly evaluating whether acute diaphragmatic breathing can improve maximal dynamic strength outcomes such as 1RM performance.

Beyond resistance training, diaphragmatic breathing interventions have been used in athletic contexts outside of maximal strength, with proposed benefits including reduced perceived stress, enhanced focus, and improved autonomic regulation—all mechanisms that may influence motor execution under pressure (3). In addition, there is an existing rationale for heavy lifting because increases in intra-abdominal pressure, driven in part by

the diaphragm and abdominal coactivation, can increase lumbar stiffness and contribute to overall trunk stability during high-load tasks (4). These psychophysiologic and mechanical pathways provide a basis to test whether a brief diaphragmatic breathing warm-up acutely improves 1RM deadlift performance.

Optimizing warmup strategies is a priority for strength and conditioning professionals, because acute interventions that enhance readiness can yield meaningful improvements in performance. Traditional warm-ups typically emphasize progressive loading, mobility, or muscle activation (12), but targeted breathing protocols may offer an additional mechanism to improve trunk stiffness and force production capacity. If effective, diaphragmatic breathing would represent a practical and low-cost warm-up intervention that could be readily implemented in applied training and competition settings. Therefore, the purpose of this study was to investigate the acute effect of a diaphragmatic breathing warm-up on maximal deadlift performance.

Methods

Experimental Approach to the Problem

This study used a randomized crossover design to investigate the acute effects of diaphragmatic breathing on maximal deadlift performance. Each subject completed 2 testing conditions: 1 with a diaphragmatic breathing intervention and 1 without (control). The order of conditions was randomized, and sessions were separated by a minimum of 48 hours. All testing was conducted at the same facility under similar environmental conditions and at approximately the same time of day to control for circadian effects. The primary dependent variable was 1RM deadlift load.

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Subjects

Forty resistance-trained adults (32 men, 8 women; age 24.8 ± 6.8 years; height 175.4 ± 8.5 cm; body mass 83.8 ± 17.0 kg) completed all testing sessions. All subjects met the minimum inclusion criteria of at least 3 months of resistance training experience. Although formal training age was not recorded, all subjects reported experience executing the barbell deadlift and were familiar with maximal or near-maximal lifting.

Exclusion criteria included musculoskeletal injuries or conditions that would prevent safe performance of the deadlift. Each subject completed a Physical Activity Readiness Questionnaire and provided written informed consent before participation. The study protocol was approved by the Maryville University Institutional Review Board (Protocol #24-80).

Procedures

At the initial visit, subjects' height and body mass were recorded, and they were familiarized with the study procedures. Subjects were instructed to refrain from strenuous lower-body exercises for at least 48 hours before each testing session. In the control condition, subjects performed a standardized warm-up followed by a 1RM deadlift assessment.

In the experimental condition, subjects performed a 5-minute diaphragmatic breathing protocol in the 90/90 position. Subjects lay supine with hips and knees flexed to 90°, ankles supported on a bench, and a foam roller placed between the knees to provide light adduction. They were instructed to inhale forcefully through the nose and exhale slowly through the mouth as if breathing through a straw, emphasizing a long, controlled exhalation (~5 seconds). Throughout each breath cycle, subjects were cued to keep the ribs depressed, maintain a gentle pressure on the foam roller, and apply a slight hamstring contraction against the bench to raise the tailbone slightly off the floor. This position promoted coactivation of the hamstrings and abdominal musculature while aligning the ribs and pelvis to practice the neutral bracing position desirable for heavy lifting. Breathing cadence was self-paced and standardized by cues: nasal inhalation (~2–3 seconds) followed by a long oral exhalation (~5 seconds), repeated continuously for 5 minutes ($\approx 35 = 55$ total breaths). The protocol was demonstrated in a standardized instructional video and performed continuously for 5 minutes before deadlift testing.

The progression of repetition ranges and relative intensities were selected to align with National Strength and Conditioning Association (NSCA) recommendations for 1RM testing, allowing for gradual neuromuscular preparation while minimizing fatigue accumulation before maximal effort attempts. Subjects completed 5 minutes of light aerobic exercise on a treadmill or stationary bike. After the completion of the aerobic portion of the warm-up, subjects performed 5–10 repetitions at 30–40% of their estimated 1-repetition max followed by a 2-minute rest, 3–5 repetitions with 50–60% of their estimated 1-repetition max followed by a 2-minute rest, 2–3 repetitions with 80–90% of their estimated 1-repetition max followed by a 2-minute rest. After the conclusion of this warm-up protocol, the subjects were given progressive 1-repetition max attempts with 2–3% load adjustments until failure, or a maximal load was achieved (6). Chalk was permitted throughout both the warm-up and 1-repetition max attempts, but no supportive equipment (e.g., lifting straps, belts) was allowed.

A lift was classified as maximal when the subject successfully completed a repetition through the full concentric range of

motion and subsequently failed to complete a heavier attempt despite adequate rest times. Failure was defined as inability to lift the barbell from the floor or inability to complete the concentric phase even if the barbell had broken inertia from the floor.

Statistical Analyses

Normality of the 1RM difference scores was assessed using the Shapiro–Wilk test ($W = 0.907, p = 0.003$), which indicated that the assumption of normality was not satisfied. Therefore, the Wilcoxon signed-rank test was used to compare 1RM between conditions. Effect sizes were calculated using Cohen's d ($d = 0.34$), interpreted as small (<0.5), moderate ($0.5–0.79$), or large (≥ 0.8), and the matched-pairs rank-biserial correlation ($r = 0.38$), interpreted as small (0.1), medium (0.3), or large (0.5). Responder analysis was also performed to evaluate individual variability. Subjects were categorized as responders if their experimental 1RM exceeded their control 1RM by at least 6.27 kg, based on the minimal detectable change reported for the 3RM deadlift (9). Subjects who did not exceed this threshold were categorized as nonresponders. Statistical significance was accepted at $p < 0.05$. Analyses were conducted using Python (Version 3.11; Python Software Foundation, Wilmington, DE), with SciPy (1.14.1) and NumPy (1.24.0) libraries.

Results

Deadlift 1RM

Deadlift 1RM was significantly greater in the diaphragmatic breathing condition than control. The Wilcoxon signed-rank test indicated a statistically significant effect (test statistic = 233.5, $p = 0.029$). The mean difference between conditions was 2.90 ± 8.61 kg, with a median difference of 4.53 kg. Cohen's d indicated a small effect ($d = 0.34$), while the rank-biserial correlation suggested a medium effect ($r = 0.38$). Twenty-four subjects (60.0%) improved their 1RM in the experimental condition, 15 subjects (37.5%) lifted less, and 1 subject (2.5%) showed no change.

Responder Analysis

Using the 6.27 kg cut point as the minimal detectable change for deadlift performance (10), 7 subjects (17.5%) were classified as responders, exceeding this threshold with improvements ranging up to 31.75 kg. Of the other subjects, 18 (45.0%) improved their 1RM but by less than the 6.27 kg threshold. Of the 15 subjects (37.5%) who demonstrated a decrease in performance, 13 experienced small decreases of <6.27 kg, where 2 showed a larger decrease ≥ 6.27 kg. These distributions of individual responses are illustrated in Figure 1.

Discussion

The primary finding of this study was that a brief diaphragmatic breathing warm-up produced a statistically significant, though small, improvement in maximal deadlift performance compared with a controlled condition. On average, subjects lifted 2.9 kg more (+2.8%) after the intervention, with 17.50% classified as responders. These results suggest that targeted breathing strategies may acutely influence maximal strength outcomes in resistance-trained individuals.

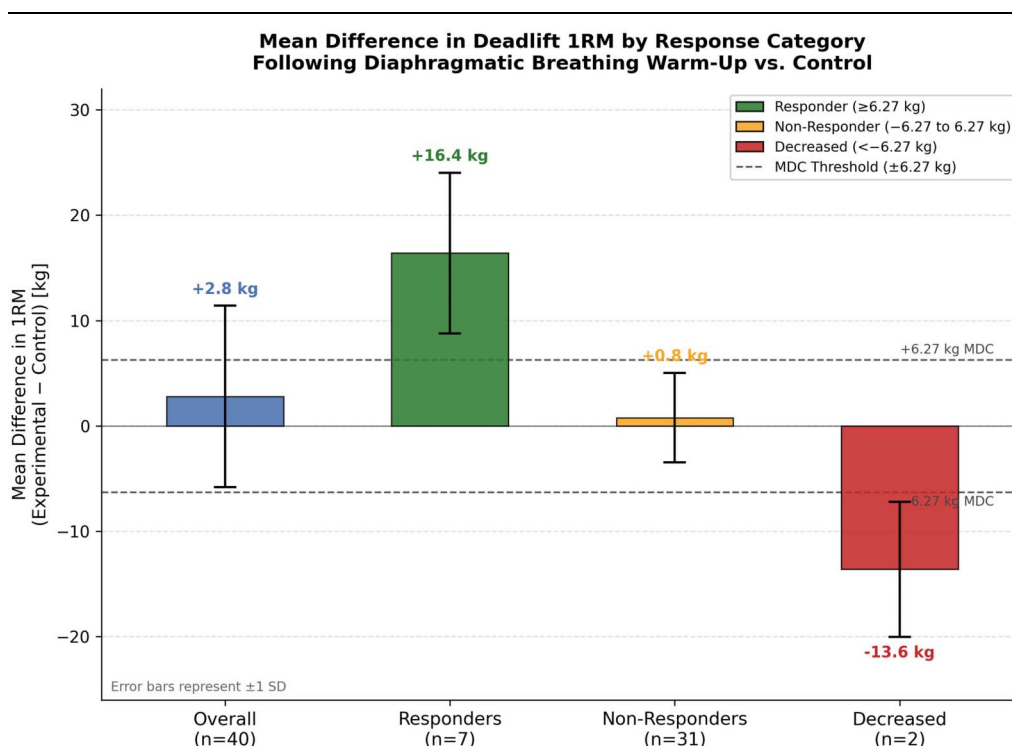


Figure 1. Mean difference in deadlift one-repetition maximum (1RM; kg) between experimental (diaphragmatic breathing warm-up) and control conditions, stratified by response category. Bars represent mean change for the overall sample ($n = 40$), responders ($\geq +6.27$ kg; $n = 7$), non-responders (-6.27 to $+6.27$ kg; $n = 31$), and decreased performers (≤ -6.27 kg; $n = 2$). Error bars indicate ± 1 standard deviation. Dashed horizontal lines denote the minimal detectable change (MDC; ± 6.27 kg). Positive values indicate greater 1RM performance following the experimental condition relative to control.

The observed improvement may be explained by the diaphragm's role in trunk stabilization and IAP regulation. The literature has highlighted the diaphragm as a critical contributor to postural control and core stability (5,7), and breathing interventions have been shown to enhance balance and neuromuscular efficiency (11). In the context of the deadlift, improved coordination of trunk stabilization and IAP generation may facilitate more efficient force transfer between the hip and shoulder complexes, thereby supporting higher maximal loads. However, because trunk stiffness and IAP were not directly measured in this study, this proposed mechanism should be interpreted with caution (1).

The importance of trunk stabilization in deadlift performance is further supported by electromyographic (EMG) research demonstrating that erector spinae muscles show greater activation than peripheral muscles during deadlifting (11). This finding suggests that trunk stability, rather than just peripheral muscle strength, is a critical determinant of deadlift performance. Diaphragmatic breathing interventions may help further optimize the contribution of these deep stabilizing muscles to force production by enhancing the coordination between the diaphragm and other trunk stabilizers.

Although the magnitude of improvement was modest, even small changes can have meaningful implications in strength testing and competition. For athletes competing in powerlifting or strength-dependent sports, a 2–3% increase in maximal load may influence rankings or qualifying totals. Furthermore, deep diaphragmatic breathing has been shown to activate the parasympathetic nervous system, promoting relaxation, improved focus, and reduced preperformance anxiety (5). These effects

could optimize attentional control and motor execution, providing a “dual benefit” not only by enhancing trunk stiffness but also placing the athlete in a more efficient psychological state before maximal exertion.

An additional explanation for the observed improvement involves the isometric activation of the posterior chain during the 90/90 setup. Pressing the heels into the bench produced a low-level hamstring curl and glute contraction, while adductor engagement from squeezing the foam roller likely recruited the adductor magnus, an important synergist for hip extension. This coactivation may have enhanced neuromuscular readiness of the posterior chain before lifting, thereby contributing to the observed performance benefit independent of diaphragmatic mechanics. Future studies using EMG or direct force assessments could help determine the relative contributions of trunk stabilization and posterior-chain muscle activation in this context.

The variability in individual responses also warrants consideration. Although 60% of the subjects improved, 40% showed no change or decreased performance. This noticeable heterogeneity is consistent with prior EMG research demonstrating interindividual differences in muscle recruitment during deadlifting (8). This may also reflect differences in breathing proficiency (e.g., ability to achieve diaphragmatic expansion while maintaining alignment of the ribs and pelvis) and pre-established bracing strategies. In addition, for some lifters, acute downregulation immediately before an 1RM attempt may be beneficial, whereas others may require higher arousal to optimize intent and force expression (9). Future work should assess breathing competency and perceived arousal to identify predictors of responsiveness.

The practical applications of this study extend beyond the performance sector, with many clinical applications as well. Diaphragmatic breathing interventions have been used to improve balance, trunk stability, and movement efficiency in rehabilitation settings (8). Incorporating structured breathing into training or recovery may, therefore, serve dual purposes: acutely enhancing performance while also supporting long-term injury prevention and functional stability.

Several limitations should be acknowledged. First, this study did not collect reliability data for the deadlift 1RM. However, a strength of the design was the use of a randomized, crossover approach, which helped control day-to-day variation in maximal performance. In addition, recruiting resistance-trained individuals likely increased the reliability of 1RM testing, because previous research has shown greater consistency in trained populations (10). Second, IAP and trunk muscle activity were not directly measured, preventing definitive confirmation of the proposed mechanisms. Third, the diaphragmatic breathing intervention involved an isometric lower-body and trunk activation through the 90/90 position, heel pressure into the bench, and adductor engagement through the foam roller. As a result, it is not possible to determine whether the observed performance changes were attributable fully to the breathing strategy, the isometric positioning, or the combination of both. Future work should examine longitudinal adaptations, include mechanistic measures such as IAP and EMG, explore whether integrating breathing practice with resistance training produces additional benefits, and include a positioning-only control condition to isolate the independent contribution of diaphragmatic breathing from neuromuscular priming associated with the setup position itself.

Practical Applications

A short diaphragmatic breathing exercise produced a statistically significant improvement in deadlift 1RM, averaging ~3 kg (median ~4.5 kg), with small-to-medium effect sizes. Given the minimal time requirement and the need for only basic equipment (a bench or box and a foam roller), this protocol may be considered as a practical addition to warm-up routines. However, improvements were not universal, with 60% of subjects showing gains and 40% showing no change or decreases. Coaches and athletes should, therefore, view diaphragmatic breathing as a low-cost, low-risk strategy that may provide modest performance benefits for certain individuals. The duration of any performance effect is currently unknown because 1RM testing began immediately after the 5-

minute intervention. Future research should quantify the persistence across typical powerlifting or strength training timelines (e.g., through warm-ups and 3 maximal attempts and/or working sets). From an applied standpoint, a 5-minute protocol may be feasible if used in place of other warm-up components, but the time constraints in scholastic settings may favor shorter bouts of breathing (e.g., 1–3 minutes) pending empirical validation.

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